NAIROBI SCHOOL 2025 PREDICTION

KCSE ANTICIPATION EXAM

233/1

CHEMISTRY

PAPER 1 (THEORY)

TIME: 2 HOURS

NAME……………………………………………………………………………………

SCHOOL……………………………………………… **SIGN**………………………

INDEX NO…………………………………………… **ADM NO**……………………

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO CANDIDATES

- *Write your name, admission number, class, date and then sign in the spaces provided.*
- *Answer **all** questions in the spaces provided.*
- *KNEC mathematical tables and silent electronic calculators **may** be used for calculations.*
- *All workings **must** be clearly shown where necessary.*

FOR EXAMINERS USE ONLY

Questions	Maximum Score	Students Score
1-28	80	

1. a) Name another gas, which is used together with oxygen in welding**.** **(1 mark)**

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(b). Explain the change in mass that occurs when the following substances are separately heated in open crucibles.

(i) Copper metal. **(1mark)**

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(ii) Copper (II) nitrate. **(1mark)**

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2. Aluminum metal is a good conductor and is used for overhead cables. State any other two properties that make aluminum suitable for this use. **(2marks)**

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3. Give two reasons why helium is used in weather balloons**.** **(2marks)**

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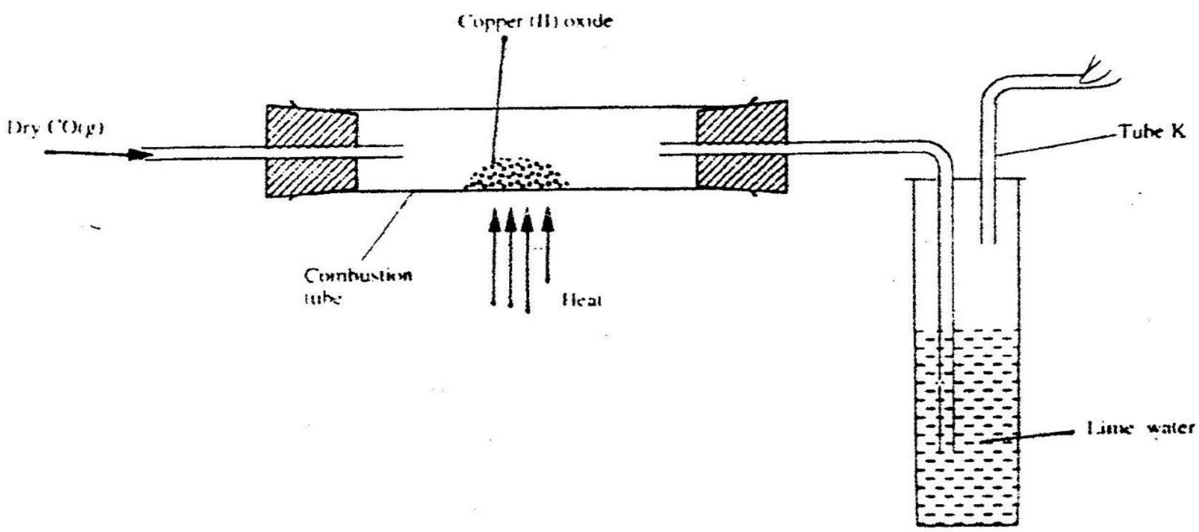
4. Draw two positional isomers of the third member of alkyne series. **(3marks)**

5. The apparatus shown below was used to investigate the effect of carbon (ii) oxide on copper (II) oxide.



a) State the observation that was made in the combustion tube at the end of the experiment.**(1mk)**

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b) Write an equation for the reaction that took place in the combustion tube **(1mark)**

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C) Why is it necessary to burn the gas coming out of tube K?

(1mark)

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6.Give a reason why

(i) Phosphorus is stored under water**.**

(1mark)

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ii) Chlorine gas is prepared in the fume chamber. **(1mark)**

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iii) Concentrated sulphuric acid is not used to dry ammonia gas**.**

(1mark)

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7. A certain matchstick head contains potassium chlorate and Sulphur. On striking, the two substances react to produce Sulphur (iv) oxide and potassium chloride. State the environmental effect of using such matches in large numbers. **(2marks)**

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8. When a sample of concentrated sulphuric acid was left in an open beaker in a room for two days, the volume was found to have increased slightly.

a) What property of concentrated sulphuric acid was being investigated. **(1mark)**

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b) State one use of concentrated sulphuric acid that depends on the property named above.**(1mark)**

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9. The following two tests were carried out on chlorine water contained in two test tubes

a) A piece of colored flower was dropped into the first – tube. Explain why the flower was bleached

(2marks)

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b) The second test- tube was corked and exposed to sunlight after a few days, it was found to contain a gas that rekindled a glowing splint. Write an equation for the reaction which produced the gas.

(1mark)

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10. In the Haber process, the optimum yield of ammonia is obtained when a temperature of 450^0C, a pressure of 200 atmospheres and iron catalysts are used

$N_2 (g) + 3H_2 (g) \rightleftharpoons 2NH_3 (g); \quad \Delta H = -92kJ.$

a) How would the yield of ammonia be affected if the temperature was raised to 600^0C? Explain. **(2marks)**

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b) Give two use of ammonia.

(1mark)

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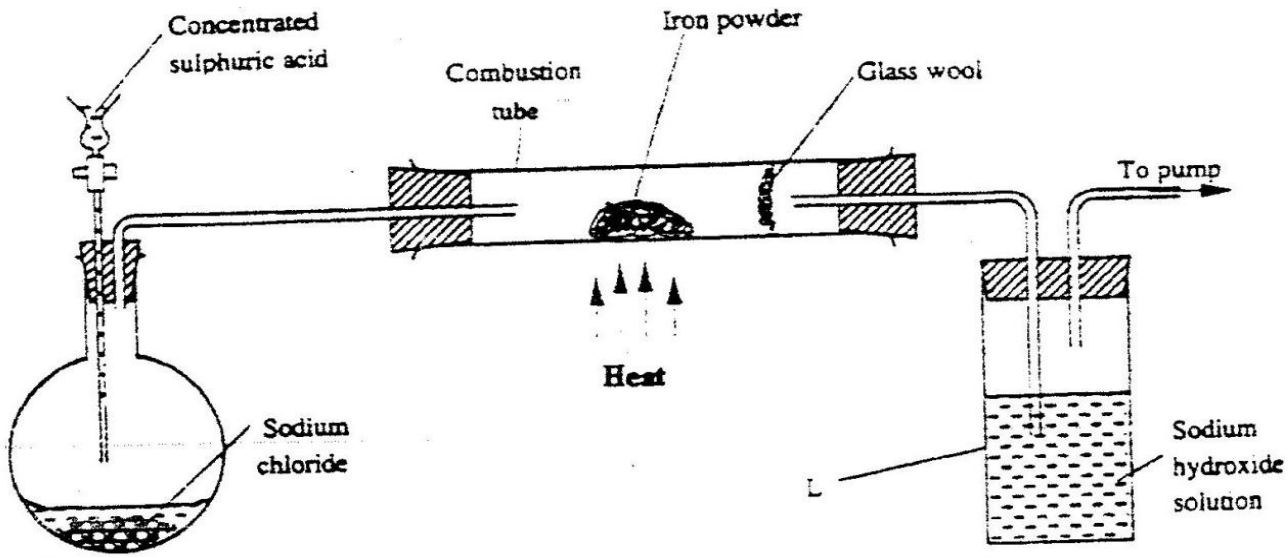
11. The set – up below was used to prepare hydrogen chloride gas and react it with iron powder. Study it and answer the questions that follow.



At the end of the reaction, the iron powder turned into a light green solid.

a) Identify the light green solid.

(1mark)

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b) At the beginning of the experiment, the pH of the solution in container L was about 14. At the end, the pH was found to be 2. Explain.

(2marks)

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12. Below is part of the Thorium decay series.

$$^{232}_{90}Th \xrightarrow{(i)} {}^{228}_{88}Ra \xrightarrow{(ii)} {}^{228}_{89}Ac \xrightarrow{(iii)} {}^{228}_{90}Th \xrightarrow{(iv)} {}^{224}_{88}Ra$$

(i) Write an overall nuclear equation for the conversion of Th-232 to Ra-224. **(1mark)**

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(ii) Give any **two** uses of radio isotopes in medicine. **(2marks)**

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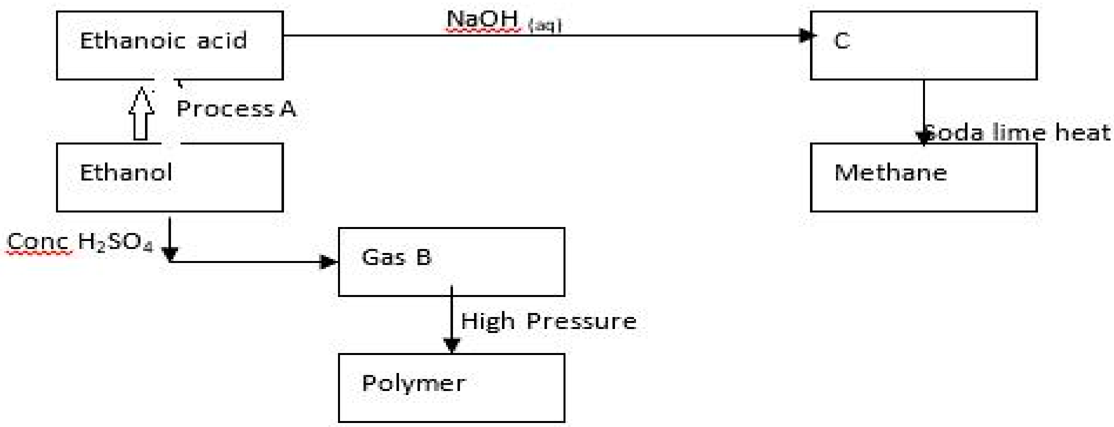
13. The flow chart below shows a series of reactions starting with ethanol , Study it and answer the questions that follow.



i) Name:

I. Process A…………………………………………………… **(1mark)**

II. Substances B and C

B.……………………………………………………………………………… **(½mark)**

C………………………………………………………………… **(½mark)**

ii) Write the equation for the reaction leading to formation of methane. **(1mark)**

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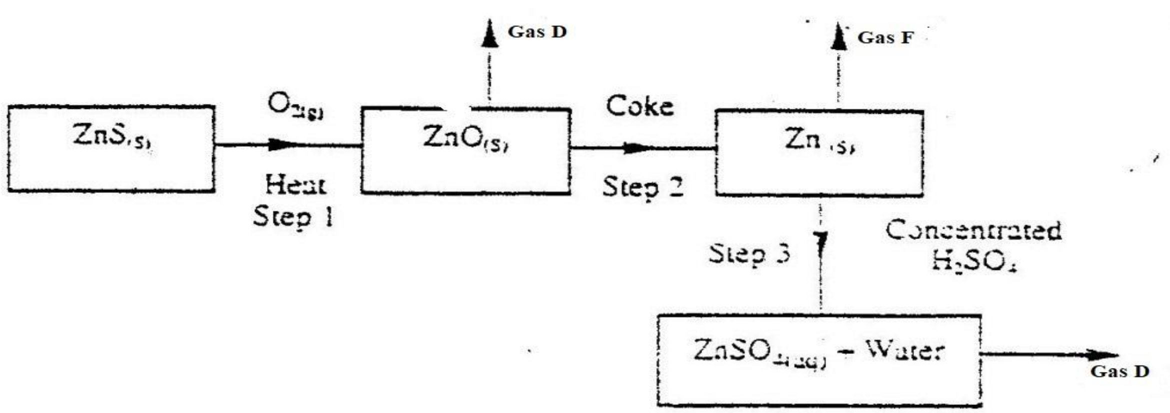
14. Study the flow chart below and answer the questions that follow.



a) State the condition necessary for the reaction in step 2 to **occur**

(1mark)

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b) Name gases D and F: **(1mark)**

i) Gas D………………………………………………

Gas F ………………………………………………..

ii) State one use of zinc metal.

(1mark)

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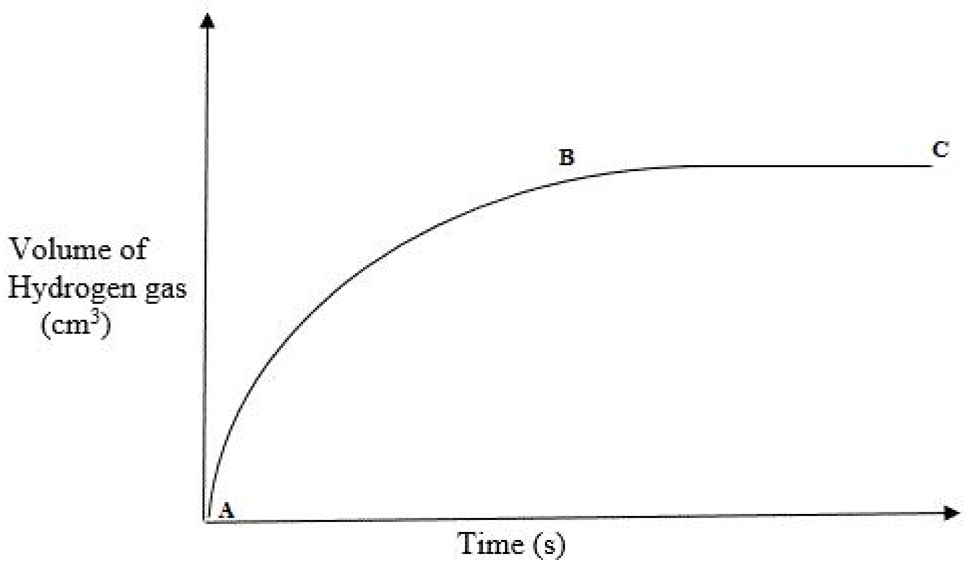
15. The reaction between a piece of magnesium ribbon with excess 2M hydrochloric acid was investigated at 25°C by measuring the volume of hydrogen gas produced as the reaction progressed. The sketch below represents the graph that was obtained.



a) Explain the shape of the curve between B and C. **(1mark)**

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b) Suggest another parameter that can be used to determine the rate of the above reaction **(1mk)**

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c) On the same diagram, sketch the curve that would be obtained if the experiment is repeated using powdered magnesium metal.
(1mark)

16. Zinc oxide reacts with acids and alkalis.

a) Write the equation for the reaction between zinc oxide and:

i) Dilute sulphuric acid **(1mark)**

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ii) Sodium hydroxide solution **(1 mark)**

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b) What property of zinc oxide is shown by the reactions in (a) above? **(1 mark)**

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17. **0.84 g** of aluminum reacted completely with chlorine gas. Calculate the volume of chlorine gas used (Molar gas volume is $24dm^3$, al = 27). **(3 marks)**

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18. Proper use of soaps in washing hands has proven to control the spread of corona novel virus.

a) Write the formula of the grey insoluble substance left in the washing basin when one uses soap with tap water given that the formula of the soap is $C_{17}H_{35}COONa$. **(1mark)**

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b) State two advantages of Soapy detergents over soapless detergents.
(2marks)

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19. a) Use the information given below to draw a labeled diagram of an electrochemical cell that can be constructed to measure the electromotive force between G and J.

$G2+(aq) + 2e \rightarrow G(s)$; $E^\theta = - 0.74$ V

$J2+(aq)+ 2e \rightarrow J(s)$; $E^\theta = -0.14$ V **(2marks)**

b) Calculate the E^θ value for the cell constructed in (a) above.

(1mark)

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20. a) Sate Le' Chetelier's principle. **(1mark)**

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b) Study the following equilibrium reaction and answer the questions that follow: -

$$\underbrace{HL_{(aq)}}_{Red} + H_2O_{(l)} \rightleftharpoons \underbrace{H_3O^+(aq)\ L^-(aq)}_{Blue}$$

Given that in an acid solution, $H_3O^+_{(aq)}$ act in place of hydrogen ions, H^+, according to the equation.

$$\mathbf{H_3O^+_{(aq)} + OH^-_{(aq)} \rightarrow 2H_2O_{(l)}}$$

Explain what would be observed when potassium hydroxide solution is added to the above equilibrium mixture. **(2marks)**

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21. The table below gives information on four elements K, L, M and N. Study it and answer the questions that follow. The letters do not represent the actual symbols of the elements.

Element	Electron arrangement	Atomic radius (nm)	Ionic radius(nm)
K	2, 8, 2	0.136	0.065
L	2, 8, 7	0.099	0.181
M	2, 8, 8, 1	0.203	0.133
N	2, 8, 8, 2	0.174	0.099

a) Which two elements have similar chemical properties? Explain. **(2marks)**

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b) Which element is a non-metal? Explain. **(1mark)**

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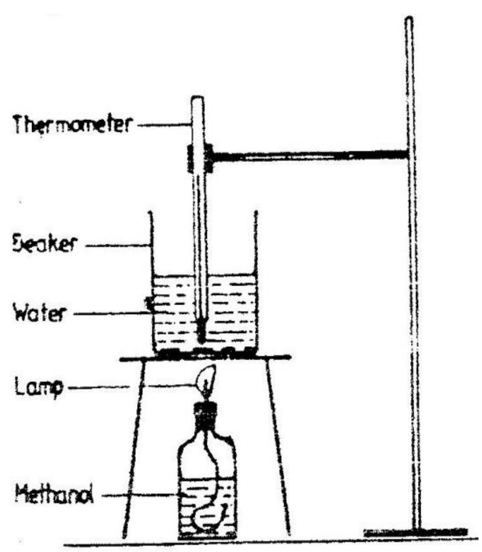
22.In an experiment to determine the heat of combustion of methanol, a student used the set-up below.



Volume of water = 500cm^3

Final temperature of water= 27.0^{0}C

Initial temperature of water = 20.0^{0}C

Final mass of lamp + methanol = 22.11g

Initial mass of lamp+ methanol= 22.98g

Density of water = 1.0g/cm^3

Specific heat capacity = 4.2kJ/g/k

Calculate:

(i) The number of moles of methanol used in this experiment given that the R.F.M is 32.

(1mark)

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(ii) The heat of combustion per mole of methanol. **(2mark)**

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23. Using dots (.) and crosses (x) to represent outermost electrons, draw diagrams to show the bonding in, CO_2 and H_3O^+. (Atomic numbers; H = 1.0,C= 14.0, O = 8)

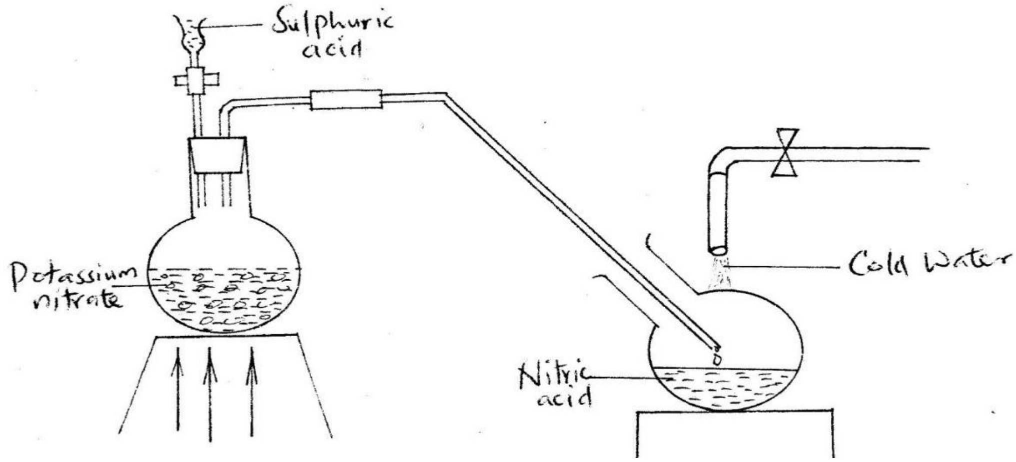
i) CO_2.

(1mark)

ii) H_3O^+

(2marks)

24. The diagram below shows a set- up that was used to prepare and collect a sample of nitric (v) acid



HEAT

a)Give a reason why it is possible to generate nitric (v) acid from sulphuric(vi) acid in the set – up. **(1mark)**

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b) Name another substance that can be used instead of potassium nitrate. **(1mark)**

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c) Give two use of nitric (v) acid. **(1mark)**

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25. When a hydrocarbon was completely burnt in oxygen, 4.2g of carbon (IV) oxide and 1.71 g of water were formed. Determine the empirical formula of the hydrocarbon

(H= 1.0 ; C=12.0 ; 0 = 16.0). **(3marks)**

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26. Starting with 50 cm^3 of 2.8M sodium hydroxide describe how a sample of pure sodium sulphate crystals can be prepared. **(3 marks)**

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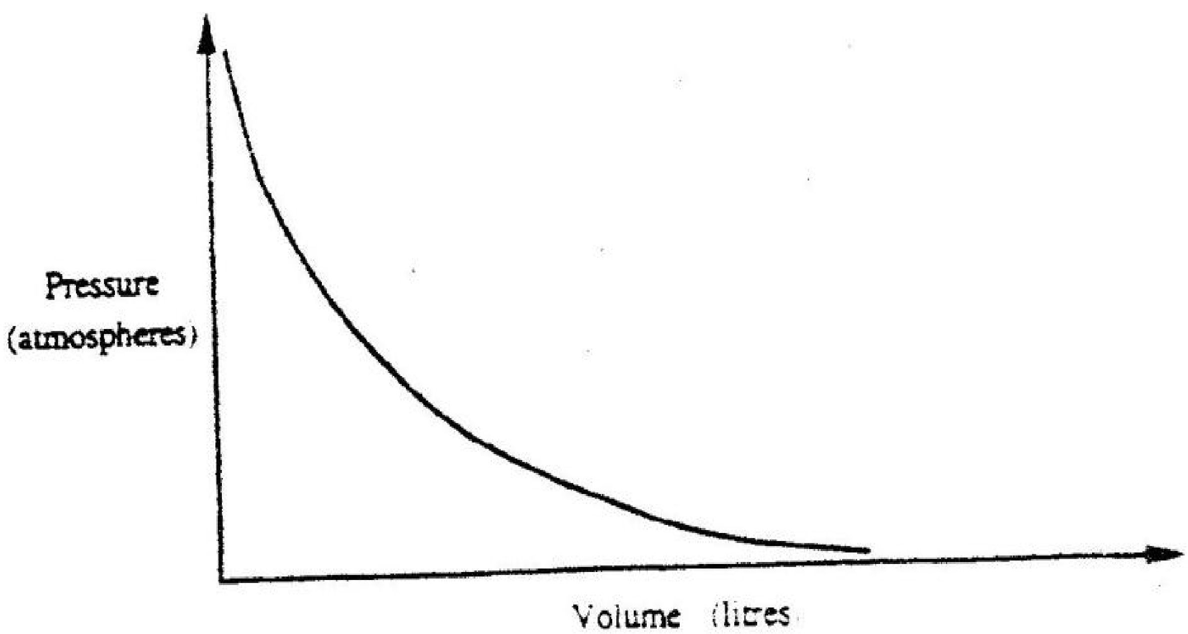
27. The graph below shows the behavior of a fixed mass of a gas at constant temperature.



a) What is the relationship between the volume and the pressure of the gas**?** **(1mark)**

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b) Three litres of oxygen gas at one atmosphere pressure were compressed to two Atmospheres at constant temperature. Calculate the volume occupied by the oxygen gas. **(2marks)**

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28. Explain the following observations

i) Very little amount of hydrogen gas is collected when dilute sulphuric acid react with calcium metal. **(1mark)**

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ii) When hydrogen chloride gas is dissolved in water, the solution turns blue litmus paper to red, while when hydrogen chloride gas is dissolved in methyl benzene; the resulting solution has no effect on the blue litmus paper. **(2marks)**

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